

a suitable location. In September 1900, the Wright brothers pitched camp at Kitty Hawk and assembled their first glider.

Testing the gliders

In many ways, this biplane resembled the machine in which they would eventually achieve powered flight. The pilot lay face down in a gap in the lower wing, a position that minimized drag. Sticking out in front of him was a movable elevator with which, using a hand lever, he controlled horizontal pitch. The wing-warping mechanism was operated by the pilot's feet. When the weather was right for a glide, the contraption was dragged to the top of a high dune. Either Wilbur or Orville climbed on board, while the other brother and a local assistant – usually the helpful Dan or Bill Tate – each held a wingtip. When the pilot was ready, they ran the glider downhill into the wind until it lifted off into skimming flight. To their immense satisfaction, the Wrights found that the controls operated well, achieving balanced flight and smooth landings.

Back in Dayton, through the winter of 1900–01, they worked on a new glider that would be the largest anyone had yet flown. The most unsatisfactory feature of the 1900 glider had been the lift, which had fallen short of expectations. The Wrights had worked out the wing size and

“We could not understand that there was anything about a bird that could not be built on a larger scale.”

ORVILLE WRIGHT

shape – crucially, the degree of camber – in line with Lilienthal's published calculations of lift and load. Now they had a second try, based on the same figures, almost doubling the surface area of the wings and using a deeper camber.

The Wrights returned to the North Carolina dunes for a second, more prolonged stay in the summer of 1901, this time setting up at Kill Devil Hills some miles from Kitty Hawk. They had by then become established members of the scattered fraternity of flight enthusiasts and were in regular correspondence with its expert, the veteran Octave Chanute, who came to witness their new round of experiments. These did not go smoothly. It took many attempts before the glider would fly at all, and when it did get off the ground, the nose

proved liable to pitch dangerously upward or downward. Substantial changes to the shape of the wing restored control, but the risks of these experiments were becoming very apparent. On one occasion Wilbur suffered a stall reminiscent of the one

that killed Lilienthal. Fortunately the front elevator proved an excellent safety device, producing a cushioned fall instead of a fatal crash.

One of the Wrights' objectives in the 1901 flights was to achieve controlled banked turns using the wing-warping mechanism, but these experiments led to a side-slipping crash in which Wilbur was injured. The brothers struggled to understand why their turns would not work, eventually deciding that the wing warping was creating drag effects that upset the machine's aerodynamics. Wilbur later wrote: “When we left Kitty Hawk at the end of 1901... we considered our experiments a failure.”

UNSTABLE BUT RESPONSIVE

Wilbur (left) and Orville (at the controls) launch their No. 3 glider, on which they tested and mastered their control systems, in 1902. A movable rear rudder worked with wing warping to give the craft lateral control, allowing smooth banked turns.



But the brothers did not give up. Instead, deciding that “the calculations on which all flying machines had been based were unreliable,” they set out to produce their own data for the lift created by various wing shapes moving at different speeds and angles. The results of these experiments with a homemade wind tunnel were then applied to a completely new glider, ready for the 1902 flying season.



AN IMPORTANT VISITOR

The picture above shows the Wrights' camp at Kill Devil Hills, North Carolina, during the visit of Octave Chanute (seated second from left) in August 1901. A pioneer of glider design, Chanute took much interest in the Wrights' glider experiments (right).

